

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460893.7063 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.044
Transit depth (Rp/Rs)^2	1.49 +/- 1.08 [%]
Orbital Inclination (inc)	83.22 +/- 2.97
Airmass coefficient 1 (a1)	1.125 +/- 0.059
Airmass coefficient 2 (a2)	-0.09 +/- 0.04
Scatter in the residuals of the lightcurve fit is	5.14 %
Best Comparison Star	#3 - [515, 161]
Optimal Aperture	2.92
Optimal Annulus	9.78
Transit Duration (day)	0.044 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.122 $\pm$ 0.044 vs 0.1326 (deviation 0.24 σ)
Duration fit vs published	0.044 d vs 0.0737 d (deviation 0.83 σ)
Mid-time O-C	21.7 min
Depth SNR	2.8
Residual scatter	5.14 %

Light curve

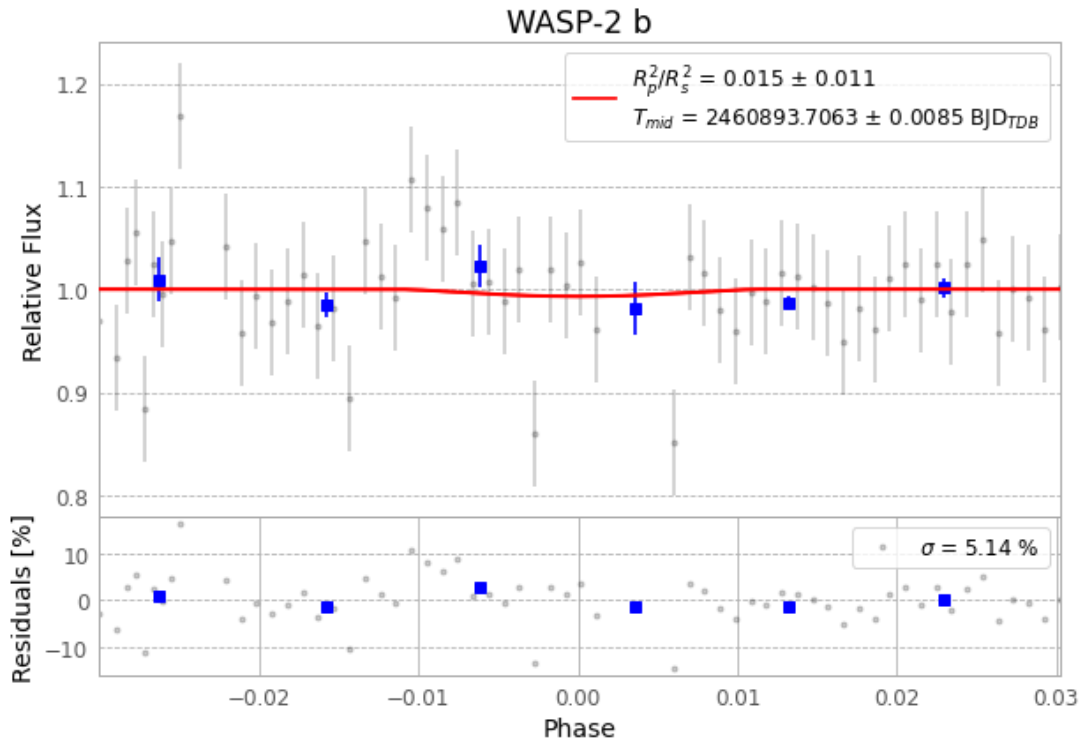


Figure 1 — FinalLightCurve_WASP-2 b_2025-08-05.png (SHA-256 eb5b7d0ec575a0ad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [321, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 52fa699693db0c51...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

65 FITS frames (43 MB), WASP-2250806031442.FITS → WASP-2250806062115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-250806.log (1b4b6a7f7946...), FinalLightCurve_WASP-2 b_2025-08-05.png (eb5b7d0ec575...), FinalLightCurve_WASP-2 b_2025-08-05.csv (977ea287b7fc...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.